BAYESIAN POOLING

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MOTIVATION: NONSTATIONARY MODELS

Nonstationary models do not fit the classical regionalization methods:

- a trend adds a coefficient at every station: $\mu_{s,t} = \alpha_s + \beta_s X_t$
- index flood and region of influence average stationary L-moments, which do not extend easily to covariates

How can we share information between similar locations in a **principled** way?

MOTIVATION: ESTIMATING NONSTATIONARY MODELS

Note

Figure shown in class: Return levels at Little Sugar Creek from a stationary model (gray) and from nonstationary models (blue), with bootstrap 95% intervals; the nonstationary intervals are far wider, especially at long return periods (Serinaldi & Kilsby, 2015).

CASE STUDY: STATIONS NEAR HOBBY AIRPORT

- The 22 NOAA Atlas 14 stations within 25 miles of Hobby Airport, from the overview.
- Data: 1-day annual maximum rainfall, 1435 station-years, with records of 29 to 125 years.
- Goal: a 100-year estimate at each station.
- Each pooling option below fits a different model to the same data.

FOUR WAYS TO POOL

$$y_{s,t} \sim \text{GEV}(\mu_{s,t}, \sigma_{s,t}, \xi_{s,t}), \quad \mu_{s,t} = \alpha_s + \beta_s X_t$$

- $y_{s,t}$: annual maximum at station s in year t; X_t: a covariate.
- For each parameter, choose how much the stations share (today's fits are stationary, $\beta_s = 0$):

1. No pooling: fit every station alone.
2. Full pooling of every parameter: one distribution for all.
3. Full pooling of some: share some, often the shape.
4. Partial pooling of some: draw station values from a common distribution.

NO POOLING

$$y_{s,t} \sim \mathrm{GEV}(\mu_{s,t}, \sigma_{s,t}, \xi_{s,t})$$

- Every parameter is fit at each station from its own record.
- One model, run once per station.
- Maximum likelihood fits give the 12.2 to 37.4 inch range of 100-year levels from the overview.

```julia
# y: one station's annual maxima
@model function no_pooling(y)
    μ ~ Normal(5, 5)
    σ ~ LogNormal(0, 2)
    ξ ~ Uniform(-0.5, 0.5)
    y .~ GeneralizedExtremeValue(μ, σ, ξ)
end
```

FULL POOLING OF EVERY PARAMETER

$$y_{s,t} \sim \mathrm{GEV}(\mu_t, \sigma_t, \xi_t)$$

- Every station has the same distribution.
- Same code as no pooling, fed every station's data.
- Longer records count for more.
- A maximum likelihood fit to all 1435 maxima gives a 100-year level of 15.2 inches at every station.

```
# y: every station's annual maxima, stacked
@model function full_pooling(y)
    μ ~ Normal(5, 5)
    σ ~ LogNormal(0, 2)
    ξ ~ Uniform(-0.5, 0.5)
    y .~ GeneralizedExtremeValue(μ, σ, ξ)
end
```

FULL POOLING OF SOME PARAMETERS

$$y_{s,t} \sim \mathrm{GEV}(\mu_{s,t}, \sigma_{s,t}, \xi_t)$$

- Like regional shape estimation.
- A maximum likelihood fit gives 100-year levels from 13.3 to 19.9 inches.

```
# station[i]: the station (1 to N) of y[i]
@model function shared_shape(y, station, N)
    μ ~ filldist(Normal(5, 5), N)
    σ ~ filldist(LogNormal(0, 2), N)
    ξ ~ Uniform(-0.5, 0.5)
    y ~ arraydist(GeneralizedExtremeValue.(
        μ[station], σ[station], ξ))
end
```

PARTIAL POOLING OF SOME PARAMETERS

$$y_{s,t} \sim \text{GEV}(\mu_{s,t}, \sigma_{s,t}, \xi_t), \quad \mu_{s,t} = \alpha_s + \beta_s X_t$$

$$\alpha_s \sim \text{Normal}(\bar{\alpha}, \omega_\alpha), \quad \beta_s \sim \text{Normal}(\bar{\beta}, \omega_\beta)$$

- $\bar{\alpha}$ and ω_α are the mean and spread of the station intercepts; $\bar{\beta}$ and ω_β are the same for the trends.
- The shape is pooled fully: it is the hardest to estimate, and the region is small and flat.

```julia
# stationary: μ = α, so there is no β
@model function partial_pooling(y, station, N)
    ᾱ ~ Normal(5, 3)
    ω_α ~ LogNormal(0, 0.5)
    α ~ filldist(Normal(ᾱ, ω_α), N)
    σ ~ filldist(LogNormal(0, 2), N)
    ξ ~ Uniform(-0.5, 0.5)
    y ~ arraydist(GeneralizedExtremeValue.(
        α[station], σ[station], ξ))
end
```

HYPERPARAMETERS

- The model estimates $\bar{\alpha}$, ω_α, $\bar{\beta}$, ω_β from the data, so the data decide how far each station is pulled toward the regional value.
- The stationary version on the last slide has $2N + 3$ parameters for N stations: 47 for the 22 stations near Hobby, and 500 draws took about 7 seconds on my laptop.

Ask

Whose α_s is pulled further toward $\bar{\alpha}$, a 30-year station or a 120-year one?

GEV PARAMETERS AS A FUNCTION OF LOCATION

$$y_{s,t} \sim \text{GEV}(\mu_{s,t}, \sigma, \xi)$$
$$\mu_{s,t} = \alpha + \beta_1 \text{lat}_s + \beta_2 \text{lon}_s$$

- One model, fit to all stations at once.
- Week 6's covariate idea, with location in place of time.
- Five parameters, however many stations.

TWO REGIONALIZATIONS OF HOUSTON RAINFALL

Note

Figure shown in class: CO2 coefficients on the GEV location (top) and scale (bottom) at 181 Gulf Coast stations, fit one station at a time (left) and pooled (right); purple is an increase with CO2 and orange a decrease (preliminary, from a paper in preparation in my group).